Functional characterisation of the pool frog (*Pelophylax lessonae*) genome as a comparative framework for future annotation of *Pelophylax perezi*

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1. INTRODUCTION

Functional genomic resources remain limited for many non-model amphibians, despite their ecological, evolutionary and biotechnological relevance. The availability of a chromosome-level reference genome and annotation for *Pelophylax lessonae* (**RefSeq: GCF_965119305.1**) provides an opportunity to investigate its functional repertoire and establish a foundation for future comparative studies within the genus *Pelophylax*.

STUDY AIM

To perform a preliminary functional characterisation of the annotated *P. lessonae* proteome and assess its potential as a reference framework for future annotation studies in *P. perezi*.

2. MATERIALS AND METHODS

The annotated *Pelophylax lessonae* genome (NCBI RefSeq: GCF_965119305.1) was used as the reference genomic resource.

A reproducible bioinformatics workflow was implemented under Ubuntu to perform functional characterisation of the predicted proteome:

- **Sequence similarity search:** DIAMOND BLASTP against UniProtKB/Swiss-Prot.
- **Functional annotation:** association of matched UniProt identifiers with Gene Ontology (GO) terms.
- **GO classification:** Biological Process (BP), Molecular Function (MF) and Cellular Component (CC).
- **Functional profiling:** GO term frequency analysis and treemap visualisation using R.

The workflow and associated scripts were documented for reproducibility (Figure 1).

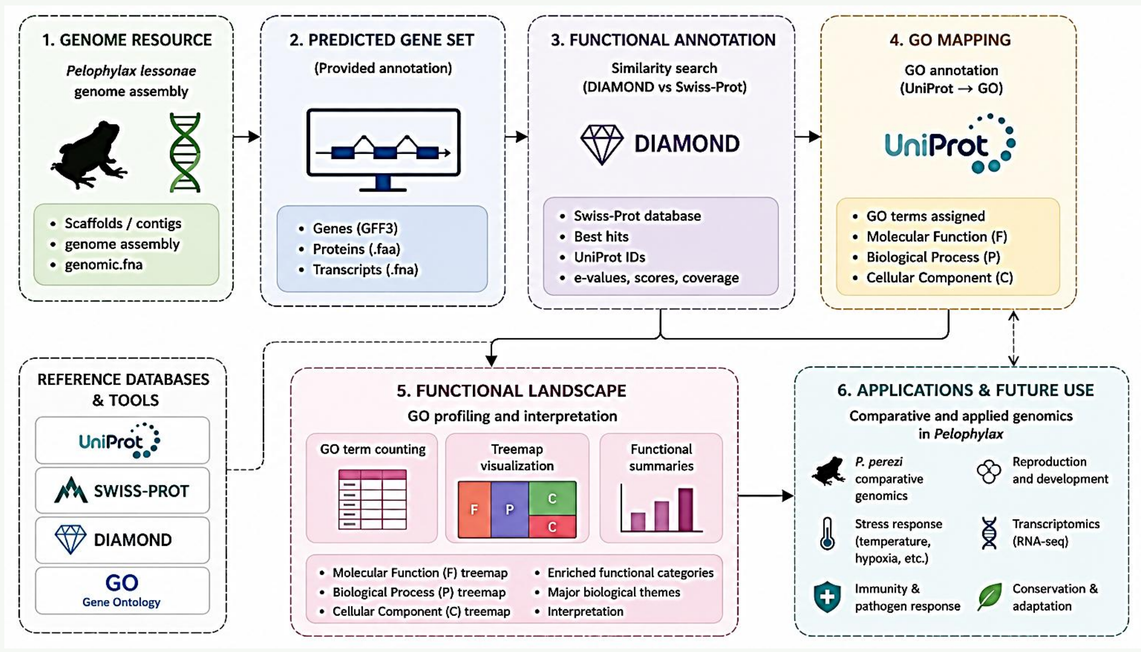


Figure 1. Reproducible workflow for functional annotation and Gene Ontology profiling of the Pelophylax lessonae predicted proteome.

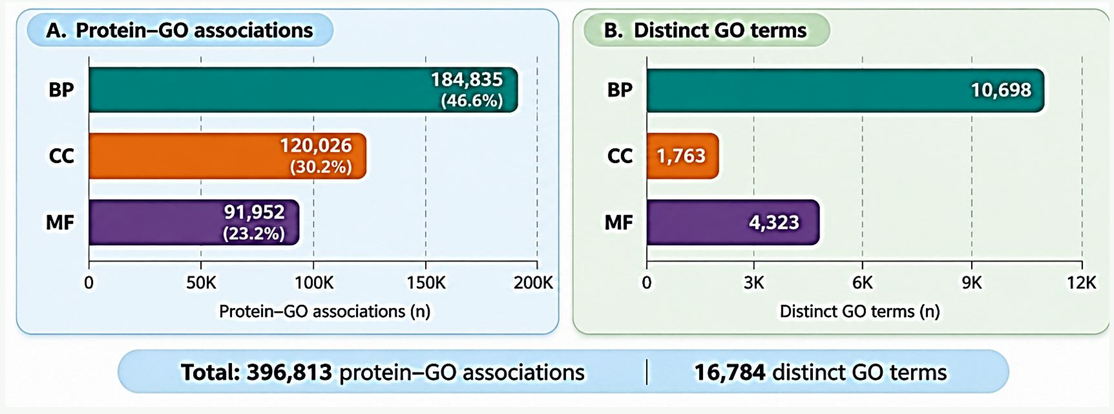


Figure 2. Quantitative summary of Gene Ontology (GO) annotations in the *Pelophylax lessonae* proteome. (A) Number and percentage of protein–GO associations across the three GO domains. (B) Number of distinct GO terms identified in each domain. BP: Biological Process; CC: Cellular Component; MF: Molecular Function. A total of 396,813 protein–GO associations and 16,784 distinct GO terms were identified.

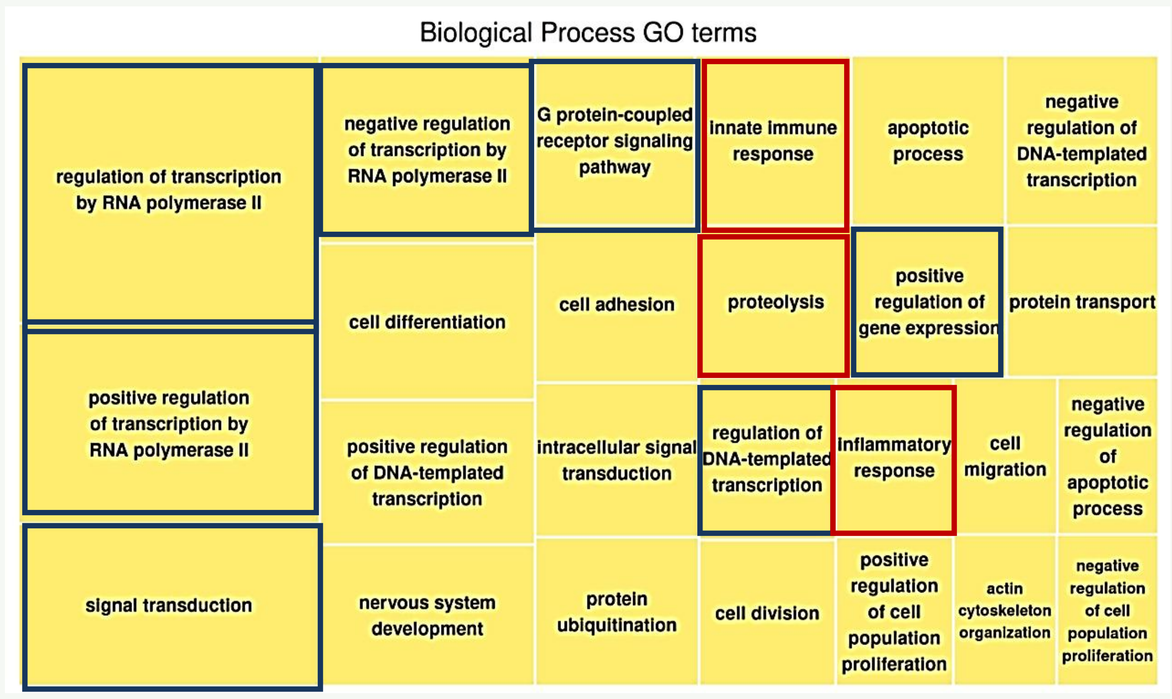


Figure 3. Biological Process Gene Ontology profile of the annotated *Pelophylax lessonae* proteome. Treemap representation of the most frequently assigned Biological Process (BP) GO terms. Rectangle areas are proportional to the frequency of protein–GO associations for each term within the displayed selection. Prominent categories include transcriptional regulation, signal transduction, cell differentiation, innate immune response and apoptosis. Functional assignments are computationally inferred and do not constitute experimental validation.

5. CONCLUSIONS

This study provides a preliminary functional characterisation of the annotated *P. lessonae* proteome, revealing a broad diversity of predicted biological functions.

The reproducible bioinformatics workflow establishes a methodological foundation for future comparative genomic studies within *Pelophylax*, particularly involving *P. perezi*.

Further analyses are needed to assess annotation accuracy and validate the framework's transferability to related species.

3. RESULTS

3.1. Quantitative functional annotation results

The functional annotation workflow generated 396,813 protein–GO associations across the three Gene Ontology domains (Figure 2).

3.2. Functional distribution of GO annotations

Gene Ontology profiling revealed a broad functional repertoire across the three GO domains: Biological Process (BP), Molecular Function (MF) and Cellular Component (CC).

Biological Process (BP) annotations revealed a broad range of predicted functional associations in the *P. lessonae* proteome. The most frequently represented GO terms were transcriptional regulation, signal transduction, cell differentiation, innate immune response, and apoptosis.

The treemap (Figure 3) highlights the relative representation of these biological processes, providing an overview of the functional diversity captured by the annotation workflow.

4. DISCUSSION

The functional annotation workflow provides a preliminary overview of the predicted functional repertoire of *P. lessonae*, revealing GO annotations associated with fundamental vertebrate biological processes.

The representation of transcriptional regulation, cellular signalling, immune response and developmental processes supports the biological plausibility of the functional assignments.

The functional annotation framework may also support future targeted searches for genes associated with innate immunity and potentially bioactive peptides, including **antimicrobial peptide (AMP)** precursors.

These results establish a starting point for future comparative genomic analyses within *Pelophylax*, particularly involving *P. perezi*.

Limitations: Functional assignments are based on sequence similarity and annotation transfer. They require further validation, and the applicability of this framework to *P. perezi* has not yet been evaluated.

REFERENCES

- Kosch TA, et al. (2025). Comparative analysis of amphibian genomes: An emerging resource for basic and applied research. *Molecular Ecology Resources*, 25(1), e14025.
- Liedtke HC, et al. (2022). Chromosome-level assembly, annotation and phylome of *Pelobates cultripes*, the western spadefoot toad. *DNA Research*, 29(3), dsac013.

DATA AVAILABILITY & REPRODUCIBILITY

Bioinformatics workflow, scripts and associated documentation:
https://github.com/Illan73/pelophylax-lessonae-functional-annotation